

10.3

此处 $\omega = 4$, $2 \cos 4t \rightarrow 2 \angle 0^\circ$ $16 \sin 4t \rightarrow 16 \angle -90^\circ = -j16$ $2H \rightarrow j\omega L = j \times 4 \times 2 = j8$ $\frac{1}{12}F \rightarrow \frac{1}{j\omega C} = \frac{1}{j \times 4 \times \frac{1}{12}} = -j3$ 在相量电路中对节点 V_0 列节点方程:

$$\frac{-j16 - V_0}{4 - j3} + 2 = \frac{V_0}{1} + \frac{V_0}{6 + j8}$$

整理得: $\frac{-j16}{4 - j3} + 2 = (1 + \frac{1}{4 - j3} + \frac{1}{6 + j8}) V_0$

$$V_0 = \frac{3.92 - j2.56}{1.22 + j0.04}$$

$$= \frac{4.682 \angle -33.15^\circ}{1.2207 \angle 1.88^\circ} = 3.835 \angle -35.02^\circ$$

$$V_0(t) = 3.835 \cos(4t - 35.02^\circ) V$$

10.32

令左上角电压为 V_0 , 右上角电压为 V_1 , 则有 $0 - V_1 = 3V_0$, 即 $V_1 = -3V_0$ 对 V_0 作 KCL:电流源向节点注入电流: $10 \angle -30^\circ$ 从节点流向地的电流为: $\frac{V_0}{2}$ 从节点 V_0 经 $j4\Omega$ 流向 V_1 的电流为: $\frac{V_0 - V_1}{j4}$ 则有 $\frac{V_0}{2} + \frac{V_0 - V_1}{j4} = 10 \angle -30^\circ$, 代入 $V_1 = -3V_0$, 得

$$\frac{V_0}{2} + \frac{V_0 - (-3V_0)}{j4} = 10 \angle -30^\circ, \text{ 可解得 } V_0 = \frac{10 \angle -30^\circ}{0.5 - j}$$

分母转为极坐标, 有 $0.5 - j = 1.118 \angle -63.435^\circ$

$$\text{即 } V_0 = \frac{10 \angle -30^\circ}{1.118 \angle -63.435^\circ} = 8.944 \angle 33.435^\circ V$$

由右侧电容阻抗为: $-j2\Omega$, $V_1 = -3V_0$,

$$\text{得 } I_0 = \frac{-3V_0}{-j2} = -j\frac{3}{2}V_0 = 1.5 \angle -90^\circ V_0$$

$$\text{即 } I_0 = 1.5 \angle -90^\circ (8.944 \angle 33.435^\circ) = 13.416 \angle -56.565^\circ A$$

10.46

① 只保留右侧直流电压源:

此时电容支路开路, 电感短路, 这时输出节点直接与 $15V$ 电源上端相连, 故 $V_1 = 15V$

② 只保留电流源:

 $\omega = 2$, $2H \rightarrow j\omega L = j4\Omega$ $\frac{1}{12}F \rightarrow \frac{1}{j\omega C} = -j6\Omega$ 则有 $6 = (\frac{1}{6} + \frac{1}{-j6} + \frac{1}{j4}) V_2$, 即 $6 = (\frac{1}{6} + \frac{j}{6} - \frac{j}{4}) V_2$

$$\text{解得 } V_2 = \frac{6}{\frac{1}{6} - \frac{j}{12}}, \text{ 化为极坐标: } V_2 = 32.18 \angle 26.56^\circ$$

$$\text{即 } V_2 = 32.18 \sin(2t + 26.56^\circ) V$$

③ 只保留左侧交流电压源:

$$\omega = 3, \quad 2H \rightarrow j\omega L = j6\Omega$$

$$\frac{1}{12}F \rightarrow \frac{1}{j\omega C} = -j4\Omega$$

对节点 V_3 列节点方程:

$$\frac{18 - V_3}{6} = \frac{V_3}{-j4} + \frac{V_3}{j6}, \quad \text{可解得: } V_3 = 16.10 \angle -26.56^\circ$$

$$\text{即 } V_3 = 16.10 \cos(3t - 26.56^\circ) V$$

由叠加定理:

$$V_o(t) = V_1 + V_2 + V_3$$

$$\text{则 } V_o(t) = 15 + 32.18 \sin(2t + 26.56^\circ) + 16.10 \cos(3t - 26.56^\circ) V$$

10.58

① 求 Z_{Th} :

原电路中, 右侧电压源置 0 后相当于短路, 即右侧支路仅剩 10Ω .

则从 $a-b$ 看去, 有两阻抗并联, 则

$$Z_{eq} = 10 \parallel (-j10) = \frac{-j100}{10 - j10} = \frac{1000 - j1000}{200}$$

$$\text{即 } Z_{eq} = 5 - j5$$

$$\text{极坐标形式为 } Z_{eq} = 7.071 \angle -45^\circ \Omega$$

$$\text{即 } Z_{Th} = Z_{eq} = 7.071 \angle -45^\circ \Omega$$

② 求 V_{Th} :

$$V_{Th} = V_{ab}, \quad \text{令 } V_{Th} = V_a, V_b = 0. \quad \text{对 } a \text{ 列节点方程, 有:}$$

$$\frac{V_a}{-j10} + \frac{V_a - j30}{10} = 3 \angle 0^\circ, \quad \text{可解得 } V_a = 30V$$

$$\text{因此 } V_{Th} = V_a = 30 \angle 0^\circ V$$

$$\text{即 等效电路 } V_{Th} = 30 \angle 0^\circ V, \quad Z_{Th} = 7.071 \angle -45^\circ \Omega$$

10.66

$$\omega = 10 \text{ rad/s}, \quad L = \frac{1}{2}H, \quad Z_L = j\omega L = j10 \times \frac{1}{2} = j5\Omega$$

$$\text{电容 } C = 10mF = 10 \times 10^{-3}F, \quad Z_C = \frac{1}{j\omega C} = \frac{1}{j \times 10 \times 10^{-3}} = \frac{1}{j \times 0.1} = -j10\Omega$$

$$2 \sin \omega t \rightarrow 2 \angle -90^\circ = -j2A,$$

$$12 \cos \omega t \rightarrow 12 \angle 0^\circ V$$

① 求 Z_N :

$$\text{令测试电流为 } I_x = 1A, \text{ 设端口电压为 } V_x, \text{ 则 } Z_N = \frac{V_x}{I_x} = V_x,$$

对端口节点列 KCL, 有:

$$1 + 2V_o = \frac{V_x}{j5} + \frac{V_x}{10 - j10}, \quad \text{代入 } V_o = \frac{10V_x}{10 - j10}, \text{ 得}$$

$$1 + 2\left(\frac{10V_x}{10 - j10}\right) = \frac{V_x}{j5} + \frac{V_x}{10 - j10}$$

$$\text{可解得: } V_x = \frac{1}{\frac{1}{j5} - \frac{19}{10 - j10}}, \quad \text{化为极坐标: } V_x = 0.67 \angle 129.56^\circ \Omega$$

$$Z_N = V_x = 0.67 \angle 129.56^\circ \Omega$$

② 求 I_N :

$$\text{因为 } I_N = \frac{V_{Th}}{Z_N}, \text{ 先求 } V_{Th}: \quad 2 \sin \omega t \rightarrow -j2A, \quad 12 \cos \omega t \rightarrow 12 \angle 0^\circ V, \quad Z_C = -j10\Omega$$

$$Z_L = j5\Omega,$$

设中间回路电流为 I , 有 $(10-j10+j5)I-10(-j2)+j5(2V_0)-12=0$,

此时 $V_0 = 10(-j2-I)$, 代入后可解得 $I = \frac{188+j20}{-10+j10.5}$,

$V_{th} = j5(I+2V_0)$, 代入得 $V_{th} = 29.79\angle-3.6^\circ \text{ V}$

则诺顿电流为 $I_N = \frac{V_{th}}{Z_N}$, 代入已求得数据得

$$I_N = \frac{29.79\angle-3.6^\circ}{0.67\angle129.56^\circ} = 44.46\angle(-3.6^\circ-129.56^\circ) = 44.46\angle-133.16^\circ \text{ A}$$

即等效诺顿电路:

$$I_N = 44.46\angle-133.16^\circ \text{ A}, \quad Z_N = 0.67\angle129.56^\circ \Omega$$